

Competitive Exam Practice 6: Singly Linked Lists

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

September 26, 2026

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type.

Q1 [*GATE CSE 1994 Q1.17, verified — GATEOverflow V2 p.233*] [**MCQ, 1 mark**] Linked lists are *not* suitable data structures for which one of the following problems?

- (A) Insertion sort
- (B) Binary search
- (C) Radix sort
- (D) Polynomial manipulation

Q2 [*GATE CSE 2002 Q1.5, verified — GATEOverflow V2 p.234*] [**MCQ, 1 mark**] In the worst case, the number of comparisons needed to search a singly linked list of length n for a given element is:

- (A) $\log n$
- (B) $n/2$
- (C) $\log_2 n - 1$
- (D) n

Q3 [*GATE CSE 2003 Q90, verified — GATEOverflow V2 pp.234–235*] [**MCQ, 2 marks**] Consider the function `f` defined below.

```
struct item {
    int data;
    struct item * next;
};
int f(struct item *p) {
    return ((p == NULL) || (p->next == NULL) ||
            ((p->data <= p->next->data) &&
             f(p->next)));
}
```

For a given linked list `p`, the function `f` returns 1 if and only if:

- (A) the list is empty or has exactly one element

- (B) the elements in the list are sorted in non-decreasing order of data value
- (C) the elements in the list are sorted in non-increasing order of data value
- (D) not all elements in the list have the same data value

Q4 [GATE CSE 2008 Q62, verified — GATEOverflow V2 p.235] [MCQ, 2 marks] The following C function takes a single-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing the integers 1, 2, 3, 4, 5, 6, 7 in the given order. What will be the contents of the list after the function completes execution?

```
struct node {
    int value;
    struct node *next;
};

void rearrange(struct node *list) {
    struct node *p, *q;
    int temp;
    if (!list || !list->next) return;
    p = list; q = list->next;
    while(q) {
        temp = p->value; p->value = q->value;
        q->value = temp; p = q->next;
        q = p ? p->next : 0;
    }
}
```

- (A) 1, 2, 3, 4, 5, 6, 7
- (B) 2, 1, 4, 3, 6, 5, 7
- (C) 1, 3, 2, 5, 4, 7, 6
- (D) 2, 3, 4, 5, 6, 7, 1

Q5 [GATE CSE 2010 Q36, verified — GATEOverflow V2 pp.235–236] [MCQ, 2 marks] The following C function takes a singly-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank.

```
typedef struct node {
    int value;
    struct node *next;
} Node;

Node *move_to_front(Node *head) {
    Node *p, *q;
    if ((head == NULL) || (head->next == NULL))
        return head;
    q = NULL; p = head;
    while (p->next != NULL) {
        q = p;
        p = p->next;
    }
```

```

    }
    -----
    return head;
}

```

Choose the correct alternative to replace the blank line.

- (A) `q = NULL; p->next = head; head = p;`
- (B) `q->next = NULL; head = p; p->next = head;`
- (C) `head = p; p->next = q; q->next = NULL;`
- (D) `q->next = NULL; p->next = head; head = p;`

Q6 [*GATE CSE 2022 Q5, verified — GATEOverflow V2 p.237*] [MCQ, 1 mark] Consider the problem of reversing a singly linked list. Which one of the following statements is *TRUE* about the time complexity of algorithms that solve the above problem in $O(1)$ space?

- (A) The best algorithm for the problem takes $\Theta(n)$ time in the worst case.
- (B) The best algorithm for the problem takes $\Theta(n \log n)$ time in the worst case.
- (C) The best algorithm for the problem takes $\Theta(n^2)$ time in the worst case.
- (D) It is not possible to reverse a singly linked list in $O(1)$ space.

Q7 [*Original, GATE-pattern*] [MCQ, 1 mark] A singly linked list holds `head` $\rightarrow$ 11 $\rightarrow$ 23 $\rightarrow$ NULL. A new node with data 4 is inserted at the front using

```
newNode->data = 4;
```

followed by two pointer assignments. Which order is safe, and what is the resulting list?

- (A) `newNode->next = head;` then `head = newNode;` giving `head` $\rightarrow$ 4 $\rightarrow$ 11 $\rightarrow$ 23 $\rightarrow$ NULL
- (B) `head = newNode;` then `newNode->next = head;` giving `head` $\rightarrow$ 4 $\rightarrow$ 11 $\rightarrow$ 23 $\rightarrow$ NULL
- (C) `newNode->next = head;` then `head = newNode;` giving `head` $\rightarrow$ 11 $\rightarrow$ 23 $\rightarrow$ 4 $\rightarrow$ NULL
- (D) `head = newNode;` then `newNode->next = head;` giving `head` $\rightarrow$ 4 $\rightarrow$ NULL with the old chain orphaned

Q8 [*Original, GATE-pattern*] [MCQ, 2 marks] A singly linked list holds `head` $\rightarrow$ 8 $\rightarrow$ 13 $\rightarrow$ 29 $\rightarrow$ NULL. The first occurrence of 13 is deleted by the deck's delete (find the predecessor, bypass, free). Afterwards, which statement is TRUE?

- (A) The list is `head` $\rightarrow$ 8 $\rightarrow$ 29 $\rightarrow$ NULL; the bypass itself is $O(1)$ worst case once the predecessor is in hand, and the overall delete is $O(n)$ worst case because of the search
- (B) The list is `head` $\rightarrow$ 8 $\rightarrow$ 13 $\rightarrow$ 29 $\rightarrow$ NULL; the bypass itself is $O(n)$ worst case
- (C) The list is `head` $\rightarrow$ 13 $\rightarrow$ 29 $\rightarrow$ NULL; deleting any value costs $O(1)$ worst case including the search
- (D) The list is `head` $\rightarrow$ 8 $\rightarrow$ 29 $\rightarrow$ NULL; the overall delete is $O(1)$ worst case because no walk is needed